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Multilayered Flexible Interconnect Circuits with Controllably Interconnectable Busbars and Methods for Operating Thereof

Abstract

Provided are multilayered flexible interconnect circuits comprising a plurality of busbars and a plurality of interconnecting units. The plurality of busbars comprises a first busbar and a second busbar. Each of the plurality of interconnecting units is connected to a first busbar and a second busbar and separately connected to an interconnecting-unit control line. Each of the plurality of interconnecting units is stacked between and connected to each of a portion of the first busbar and a portion of the second busbar. Each of the plurality of interconnecting units maintains electrical disconnection between the first busbar and the second busbar before receiving an electrical input through the interconnecting-unit control line. Each of the plurality of interconnecting units is configured to electrically connect the first busbar and the second busbar after receiving the electrical input through the interconnecting-unit control line.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application 63/563,773 (Attorney Docket No. CLNKP029P) by Tate, et. al., titled: “Multilayered Flexible Interconnect Circuits with Controllably Interconnectable Busbars and Methods for Operating Thereof”, filed on 2024 Mar. 11, which is incorporated herein by reference in its entirety for all purposes.

FIELD OF TECHNOLOGY

[0002] This patent application relates generally to busbar interconnects, and more specifically to electric vehicle battery busbar interconnects.

BACKGROUND

[0003] Battery cells in battery packs and other types of battery assemblies are typically interconnected using individual busbars. Each busbar is stamped from a sufficiently thick metal sheet (selected based on current ratings) and individually handled during the busbar installation (e.g., positioned over and welded to the battery terminals). Furthermore, additional circuits (e.g., voltage sense harnesses) can be installed and connected to these busbars during the battery pack fabrication. Overall, existing busbars do not provide re-routing of electrical connections to maintain current flow around a failing or failed battery cell within a battery pack.

[0004] What is needed is new multilayered flexible interconnect circuits comprising a plurality of busbars and a plurality of interconnecting units capable of selectably interconnecting individuals of the plurality of busbars, battery backs comprising such multilayered flexible interconnect circuits, and methods for operating thereof.

SUMMARY

[0005] Provided are multilayered flexible interconnect circuits comprising a plurality of busbars and a plurality of interconnecting units. The plurality of busbars comprises a first busbar and a second busbar. Each of the plurality of interconnecting units is connected to a first busbar and a second busbar and separately connected to an interconnecting-unit control line. Each of the plurality of interconnecting units is stacked between and connected to each of a portion of the first busbar and a portion of the second busbar. Each of the plurality of interconnecting units maintains electrical disconnection between the first busbar and the second busbar before receiving an electrical input through the interconnecting-unit control line. Each of the plurality of interconnecting units is configured to electrically connect the first busbar and the second busbar after receiving the electrical input through the interconnecting-unit control line.

[0006] Clause 1. A multilayered flexible interconnect circuit comprising: a plurality of busbars, comprising a first busbar and a second busbar, wherein each one of the plurality of busbars is defined by a contact portion, a first tab portion, and a second tab portion; a plurality of interconnecting-unit control lines, comprising an interconnecting-unit control line; and a plurality of interconnecting units, wherein: each one of the plurality of interconnecting units is stacked between and connected to each of the first tab portion of one of the plurality of busbars and the second tab portion of another one of the plurality of busbars and separately connected to an interconnecting-unit control line, each one of the plurality of interconnecting units is configured to electrically disconnect the first tab portion and the second tab portion before receiving an electrical

input through the interconnecting-unit control line, and each one of the plurality of interconnecting units is configured to electrically connect the first tab portion and the second tab portion after receiving the electrical input through the interconnecting-unit control line.

[0007] Clause 2. The multilayered flexible interconnect circuit of clause 1, wherein each one of the plurality of interconnecting units is a transistor comprising: a source electrically interconnected with either the first tab portion of the first busbar and the second tab portion of the second busbar, a drain electrically interconnected with the other of either the first tab portion of the first busbar and the second tab portion of the second busbar, and a gate electrically interconnected with the interconnecting-unit control line, wherein: each transistor is configured to reversibly electrically connect the first busbar and the second busbar after receiving the electrical input through the interconnecting-unit control line.

[0008] Clause 3. The multilayered flexible interconnect circuit of clause 1, wherein each one of the plurality of interconnecting units is a normally-open relay comprising: a first pole electrically interconnected with either the first tab portion of the first busbar and the second tab portion of the second busbar, a second pole electrically interconnected with the other of either the first tab portion of the first busbar and the second tab portion of the second busbar, and an actuation lead electrically interconnected with the interconnecting-unit control line, wherein: each normally-open relay is configured to reversibly electrically connect the first busbar and the second busbar after receiving the electrical input through the interconnecting-unit control line.

[0009] Clause 4. The multilayered flexible interconnect circuit of clause 1, wherein each one of the plurality of interconnecting units is a heat-actuated interconnector comprising: a first connecting portion electrically interconnected with either the first tab portion of the first busbar and the second tab portion of the second busbar, a second connecting portion electrically interconnected with the other of either the first tab portion of the first busbar and the second tab portion of the second busbar, a meltable portion positioned between the first connecting portion and the second connecting portion, and a heating element electrically interconnected with the interconnecting-unit control line, wherein: the heating element is configured to melt the meltable portion after receiving the electrical input through the interconnecting-unit control line and thereby irreversibly electrically connect the first busbar and the second busbar.

[0010] Clause 5. The multilayered flexible interconnect circuit of clause 4, wherein the heating element is positioned between the first connecting portion and the second connecting portion.

[0011] Clause 6. The multilayered flexible interconnect circuit of clause 4, wherein the heating element is positioned either on an opposite side of the first connecting portion from the second connecting portion or on an opposite side of the second connecting portion from the first connecting portion.

[0012] Clause 7. The multilayered flexible interconnect circuit of clause 1, wherein: each one of the plurality of busbars comprises a first conductive layer and a second conductive layer, the contact portion is formed by both the first conductive layer and the second conductive layer, stacked in the contact portion, the first tab portion is formed by the first conductive layer such that the second conductive layer does not extend to the first tab portion, and the second tab portion is formed by the second conductive layer such that the first conductive layer does not extend to the second tab portion.

[0013] Clause 8. The multilayered flexible interconnect circuit of clause 7, wherein the first conductive layer and second conductive layer are separated from each other in the contact portion by an insulation layer.

[0014] Clause 9. The multilayered flexible interconnect circuit of clause 7, wherein the first conductive layer and second conductive layer directly interface each other in the contact portion.

[0015] Clause 10. The multilayered flexible interconnect circuit of clause 9, wherein the first conductive layer and the second conductive layer directly interface each other and are welded to one another in the contact portion.

[0016] Clause 11. The multilayered flexible interconnect circuit of clause 1, wherein at least one of the plurality of interconnecting units reversibly electrically connects the first tab portion and the second tab portion after receiving the electrical input.

[0017] Clause 12. The multilayered flexible interconnect circuit of clause 1, wherein at least one of the plurality of interconnecting units is connected to one of the plurality of busbars connected to a terminal of a battery and another one of the plurality of busbars connected to another terminal of the battery.

[0018] Clause 13. The multilayered flexible interconnect circuit of clause 1, wherein at least one of the plurality of interconnecting units is connected to one of the plurality of busbars connected to a terminal of a battery and another one of the plurality of busbars connected to another terminal of another battery.

[0019] Clause 14. The multilayered flexible interconnect circuit of clause 1, wherein each of the plurality of interconnecting units irreversibly electrically connects the first tab portion and the second tab portion after receiving the electrical input.

[0020] Clause 15. The multilayered flexible interconnect circuit of clause 1, wherein the electrical input has a voltage of 1-5 volts.

[0021] Clause 16. A battery assembly comprising: a plurality of busbars comprising a first busbar and a second busbar; a plurality of interconnecting units, each connected to two of the plurality of busbars and separately connected to an interconnecting-unit control line; a plurality of battery cells comprising cell terminals; and a battery management system, wherein: each of the plurality of busbars comprises a first conductive layer and a second conductive layer, each of the plurality of busbars is defined by a contact portion, a first tab portion, and a second tab portion, the contact portion is formed by both the first conductive layer and the second conductive layer, stacked in the contact portion, the first tab portion is formed by the first conductive layer such that the second conductive layer does not extend to the first tab portion, the second tab portion is formed by the second conductive layer such that the first conductive layer does not extend to the second tab portion, the cell terminals of each one of the plurality of battery cells are each electrically connected to different ones of the plurality of busbars, each of the interconnecting-unit control lines is electronically connected with the battery management system, the plurality of interconnecting units comprise an interconnecting unit stacked between and connected to each of the first tab portion of the first busbar and the second tab portion of the second busbar, each one of the plurality of interconnecting units maintains the two of the plurality of busbars electrically disconnected from each other before receiving an electrical input through the interconnecting-unit control line, and each one of the plurality of interconnecting units is configured to electrically connect the two of the plurality of busbars after receiving the electrical input through the interconnecting-unit control line.

[0022] Clause 17. The battery assembly of clause 16, further comprising a plurality of voltage traces, wherein each of the voltage traces electrically connects one of the plurality of busbars to the battery management system.

[0023] Clause 18. The battery assembly of clause 16, further comprising a battery management system power source electrically connected to the Battery Management System, wherein the battery management system power source is at least one of the plurality of battery cells.

[0024] Clause 19. A method of operating a battery assembly, the method comprising: providing a battery pack comprising: a plurality of busbars comprising a first busbar and a second busbar; a plurality of interconnecting units, each connected to two of the plurality of busbars and separately connected to an interconnecting-unit control line; a plurality of battery cells comprising cell terminals; a plurality of voltage traces; and a battery management system, wherein: each of the plurality of busbars comprises a first conductive layer and a second conductive layer, each of the plurality of busbars is defined by a contact portion, a first tab portion, and a second tab portion, the contact portion is formed by both the first conductive layer and the second conductive layer, stacked in the contact portion, the first tab portion is formed by the first conductive layer such that

the second conductive layer does not extend to the first tab portion, the second tab portion is formed by the second conductive layer such that the first conductive layer does not extend to the second tab portion, the cell terminals of each one of the plurality of battery cells are each electrically connected to different ones of the plurality of busbars, each of the interconnecting-unit control lines is electronically connected with the battery management system, each of the voltage traces electrically connects one of the plurality of busbars to the battery management system, the battery management system is configured to determine SOH of each of the plurality of battery cells by measuring voltages at each of the plurality of busbars, the plurality of interconnecting units comprise an interconnecting unit stacked between and connected to each of the first tab portion of the first busbar and the second tab portion of the second busbar, each one of the plurality of interconnecting units maintains the two of the plurality of busbars electrically disconnected from each other before receiving an electrical input through the interconnecting-unit control line, and each one of the plurality of interconnecting units is configured to electrically connect the two of the plurality of busbars after receiving the electrical input through the interconnecting-unit control line; determining at the battery management system if the SOH of any one of the plurality of battery cells is unsatisfactory by comparing voltages measured at each of the plurality of busbars to a predetermined list of voltages; selecting at the battery management system which of the plurality of interconnecting units to provide an electrical input to; and providing, by the battery management system, an electrical input to the selected ones of the plurality of interconnecting units via at least one of the interconnecting-unit control line.

[0025] Clause 20. The method of operating a battery assembly of clause 19, the method further comprising: determining at the battery management system an SOC of any one of the plurality of battery cells with unsatisfactory SOH.

[0026] These and other embodiments are described further below with reference to the figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1A is a top view of a portion of a multilayered flexible interconnect circuit illustrating an opening in the top (second) outer insulator and various boundaries of the conductive components, in accordance with some examples.

[0028] FIG. 1B is a schematic side view of two battery cells interconnected by the busbar of a multilayered flexible interconnect circuit, in accordance with some examples.

[0029] FIG. 1C is a cross-sectional side view of a multilayered flexible interconnect circuit identifying various components and portions of the circuit, in accordance with some examples.

[0030] FIG. 1D is an expanded cross-sectional side view of the busbar support portion of the multilayered flexible interconnect circuit in FIG. 2B, in accordance with some examples.

[0031] FIG. 2A is a top view of a portion of a multilayered flexible interconnect circuit illustrating an arrangement of busbars and interconnecting units, in accordance with some examples.

[0032] FIG. 2B is a top view of a busbar formed from two conductive layers illustrating the contact portion, first tab portion, and second tab portion, in accordance with some examples.

[0033] FIG. 2C is a side view of the busbar shown in FIG. 2B, in accordance with some examples.

[0034] FIG. 2D is a side view of an interconnecting unit in contact with a first busbar and a second busbar, in accordance with some examples.

[0035] FIG. 2E is a side view of an interconnecting unit in contact with a first busbar and a second busbar, in accordance with some examples.

[0036] FIG. 2F is a side view of an interconnecting unit in contact with a first busbar and a second busbar, in accordance with some examples.

[0037] FIG. 2G is a top view of a portion of a multilayered flexible interconnect circuit comprising

a plurality of busbars and connected components, in accordance with some examples.

[0038] FIG. 3A is a schematic view of an interconnecting unit connecting a first busbar and a second busbar, illustrating the flow of electricity through the battery when no electrical input is received at the interconnecting-unit control line, in accordance with some examples.

[0039] FIG. 3B is a schematic view of an interconnecting unit connecting a first busbar and a second busbar, illustrating the flow of electricity through the interconnecting unit when an electrical input is received at the interconnecting-unit control line, in accordance with some examples.

[0040] FIG. 4A is a schematic block diagram illustrating the relationships of components general to interconnecting units, in accordance with some examples.

[0041] FIG. 4B is a schematic block diagram illustrating the relationships of components specific to an interconnecting unit comprising a transistor, in accordance with some examples.

[0042] FIG. 4C is a schematic block diagram illustrating the relationships of components specific to an interconnecting unit comprising a normally-open relay, in accordance with some examples.

[0043] FIG. 4D is a schematic block diagram illustrating the relationships of components specific to a heat-actuated interconnecting unit, in accordance with some examples.

[0044] FIG. 5 is a schematic block diagram illustrating the relationships of components of battery assemblies comprising interconnecting units, in accordance with some examples.

[0045] FIG. 6 is a process flowchart corresponding to a method of fabricating a multilayered flexible interconnect circuit, in accordance with some examples.

[0046] FIG. 7 is a process flowchart corresponding to a method of operating battery packs comprising multilayered flexible interconnect circuits with a plurality of interconnecting units.

DETAILED DESCRIPTION

[0047] In the following description, numerous specific details are outlined to provide a thorough understanding of the presented concepts. In some examples, the presented concepts are practiced without some or all of these specific details. In other examples, well-known process operations have not been described in detail to unnecessarily obscure the described concepts. While some concepts will be described in conjunction with specific examples, it will be understood that these examples are not intended to be limiting.

Introduction

[0048] As noted above, conventional busbars used for connecting battery cells are typically used as individual components stamped from thick metal sheets to ensure sufficient current capabilities. However, this individual component aspect complicates the battery pack assembly process, e.g., requiring individual handling and alignment of each component. Furthermore, these thick metal sheets may not be sufficiently flexible to accommodate various alignment deviations among battery cells, which further complicates the installation process. Finally, various additional components (besides battery cells, e.g., voltage-sense harnesses) need to be connected to busbars adding even more operational complexities.

[0049] Battery packs are collections of electrically connected battery cells. Each individual battery cell in the battery pack typically does not provide sufficient voltage or energy storage capacity to sufficiently power, for example, an electric vehicle. To achieve sufficient voltage to operate electric motors in, for example, electric vehicles, sets of batteries are often connected in series using busbars. Busbars connect the negative terminal of one battery cell to the positive terminal of an adjacent battery cell. In order to supply sufficient current, multiple series of battery cells may be connected with one another in parallel. Battery modules/packs may comprise more than 50 battery cells or even more than 100 battery cells in some designs (e.g., 800V battery packs).

[0050] A battery cell in a battery pack may fail in one of at least two failure modes during charging operations or operation of the vehicle. One possible failure mode is when the damaged battery cell acts as an open circuit between the two terminals. When one battery cell in a series fails in a failure mode resulting in an open circuit, the entire series will no longer provide power to the electric

vehicle. The battery pack may then be unable to provide sufficient power for operation of the vehicle.

[0051] If the failure mode results in a short between the terminals of one battery cell, the result may be a rapid discharge of the failed battery. Rapid discharging may result in generation of significant heat energy at the location of the battery cell. This heat generation may trigger failure of adjacent battery cells, which may also generate heat and damage further battery cells, resulting in a so-called thermal runaway event.

[0052] For some electric vehicles, it would be desirable to have a Limp-home mode available in case of failure of one or more battery cells. A limp-home mode may enable continued operation of an electric vehicle in case of a battery cell failure. A battery management system (BMS) may disconnect individual batteries from the rest of the battery pack. It may then route power from the remaining connected battery cells to allow safe operation of the vehicle but for a limited range. This may, for example, allow operation of the vehicle to a safe stopping location. It may provide sufficient power for the battery management system to continue to measure and manage the battery pack in a safe state. This may provide passengers the opportunity to park the vehicle and exit safely. It may, for example, allow operation of the vehicle to a location where the vehicle can be repaired. It may be desirable for a battery pack of a disabled vehicle to continue to provide some power for operating accessories. For example, in cold environments, it may be desirable for a disabled vehicle to operate passenger cabin heaters. Electric tractors may often be operated far from roads and repair infrastructure. It may be desirable for electric tractors to provide a limp-home mode in case of battery cell failure. In some cases, local regulations may require that an electric vehicle is equipped to provide a limp-home mode. In the case of electric aircraft powered by battery packs, it may be desirable for a limp-home mode to provide sufficient power for the vehicle to be safely landed on the ground.

[0053] In each of these applications, a battery management system may be used to both sense battery cell failure and change connections between individual battery cells. For some applications, it may be desirable to power a battery management system (BMS) with a secondary battery power source. This secondary battery power source may be, for example, a 12 V or 48 V battery pack with smaller power storage capacity than the primary battery pack. This would provide power to the BMS to continue to monitor the health of battery cells in the primary battery pack even if, after failure of one or more battery cells, the battery pack is disconnected from its primary load. However, the battery management system may itself require electrical power to continue operation. The electrical power may be provided by a backup battery system. For example, an auxiliary 12 volt or 48 volt battery pack may be provided separately to the primary battery pack. However, to reduce cost, mass, and space required to include a separate battery pack, it may be desirable to operate the BMS from the primary battery pack. In this arrangement, disconnecting the entire battery pack from powering systems of the electric vehicle in case of a battery cell failure may not be desirable. Instead, it would be desirable to disconnect a selection of batteries to allow, for example, providing continuous power to the BMS. This would require dynamic electrical connections between battery cells.

[0054] For continued operation of the vehicle and for enhanced operator safety in cases of battery cell failure, it would be desirable to limit the negative consequences of both cell failure modes on the battery pack. However, existing battery packs electrically connect batteries via static connections to busbars.

[0055] Multilayered flexible interconnect circuits described herein address various issues listed above. Specifically, a multilayered flexible interconnect circuit comprises at least two conductive layers and at least one inner insulator, which extends between these conductive layers in some circuit portions and allows conductive layers to directly interface in other circuit portions (e.g., busbar portions). In other words, when high current-carrying capabilities are needed, multiple conductive layers (e.g., all conductive layers) are in that portion of the circuit. It should be noted

that stacking multiple conductive layers increases the flexibility of this stack in comparison to a monolithic component with the same thickness (and the same current-carrying capability). Alternatively, when only low current-carrying capabilities are needed (e.g., for voltage sensing), fewer than all conductive layers (e.g., only one conductive layer) can be used in this circuit portion. Since all components of the same conductive layer are formed from the same initial metal sheet, these components may be monolithically integrated (and do not require any later connections). Furthermore, components of different conductive layers may directly interface with each other (e.g., through an opening within an inner insulator layer) and even welded to each other (e.g., through an opening within an outer insulator layer). In some examples, the components of different conductive layers may be welded to each other while welding these to various external components (e.g., battery terminals). It should be noted that one or more inner insulators allow stacking multiple conductive layers while forming electrical connections between these layers, e.g., having multiple voltage traces crossing over.

[0056] The multilayered flexible interconnect circuit comprises a plurality of busbars connecting terminals of battery cells. The multilayered flexible interconnect circuit also comprises a plurality of interconnecting units. Each interconnecting unit is connected to two of the plurality of busbars and is separately connected to an interconnecting-unit control line. Each interconnecting unit is configured to maintain the two busbars to which is connected electrically disconnected unless it receives an electrical input. Each layer interconnecting unit is configured to electrically connect the two busbars upon receiving an electrical input. Electrically connecting the two busbars changes the electrical connections among the plurality of battery cells. Electrically connecting the two busbars removes one or more battery cells from a series of battery cells while conducting electricity around the removed cells. Layer interconnecting units may comprise connected components in addition to the multilayered flexible interconnect circuit. Multiple conductive layers may be where leads for carrying current from busbars to layer interconnecting units are. Conductor layers connecting busbars to layer interconnecting units are on different conductive layers. Conductor layers connecting busbars to layer interconnecting units may have at least one insulator layer positioned between. Interconnecting-unit control lines are included in the multilayered flexible interconnect circuit. Interconnecting-unit control lines may be included on conductive layers. Interconnecting-unit control lines may, for example, carry electrical signals from the BMS to layer interconnecting units. Examples of layer interconnecting units are described in more detail below.

FIGS. 1A-1B: Examples of Multilayered Flexible Interconnect Circuit Assemblies

[0057] The functional and structural aspects of multilayered flexible interconnect circuits will now be described in the context of FIGS. 1A-1D. Specifically, FIG. 1A is an expanded top view of a portion of a multilayered flexible interconnect circuit illustrating openings **119** in the second outer insulator layer **112** and the boundaries of conductive components, in accordance with some examples. Specifically, a portion of the busbar **138** may overlap with the second outer insulator layer **112** (and, e.g., a first outer insulator layer **111** as shown in FIG. 1B) to support the busbar **138** within the multilayered flexible interconnect circuit **100**. Furthermore, FIG. 1A illustrates a voltage trace **139** extending to the busbar **138** (and being monolithic with one of the conductive layers forming the busbar **138**). A portion of the voltage trace **139** can be specifically shaped and referred to as a voltage trace connection portion **137**. The voltage trace connection portion **137** may interconnect a linear portion of the voltage trace **139** and busbar **138** and allow for the out-plane movement of the busbar **138** relative to the linear portion of the voltage trace **139** while preserving this monolithic connection. FIG. 1A also illustrates additional voltage trace **139** extending next to each other (e.g., to other busbars). As noted above, additional voltage traces can be offset along the Z-axis. Each conductive component is formed by one or more conductive layers.

[0058] FIG. 1B is a side view of battery cells **190** forming a battery assembly **180**, in accordance with some examples. In FIG. 1B, the battery cells **190** are prismatic cells, but other types of cells are within the scope. In the illustrated example, the set includes two battery cells. However, any

other number and/or arrangements of battery cells are within the scope. Each battery cell **190** comprises cell terminals **191** and **192**, such as a positive cell tab and a negative cell tab. In the illustrated example, both cell terminals **191** and **192** are positioned on the same side of battery cell **190**. However, other examples (e.g., with cell terminals being positioned on opposite sides of battery cells **190**) are also within the scope. In these examples, multiple multilayered flexible interconnect circuits may be used to interconnect the same set of battery cells **190**. The cell terminals **191** and **192** are used for interconnecting the battery cells **190**, e.g., with all cells being interconnected in series as further shown with the design of the multilayered flexible interconnect circuit **100** presented in FIG. 2A. However, other connection schemes (e.g., parallel and various combinations of parallel and in-series connections) are also within the scope.

[0059] The number of the conductive layers forming each component depends on the current carrying requirement of these specific components. Specifically, these conductive components may include busbars **138** and voltage traces **139**. Busbars **138** are examples of high-current-carrying conductive components, each formed using multiple conductive layers of the multilayered flexible interconnect circuit **100**. Busbars **138** are connected (e.g., welded) to the cell terminals **191** and **192** during the fabrication of a battery assembly **180** (e.g., a battery pack). It should be noted that during this fabrication operation, all busbars **138** are integrated and supported within the multilayered flexible interconnect circuit **100** thereby eliminating the need to handle and align each busbar (in comparison to conventional methods).

[0060] Voltage traces **139** are examples of low-current carrying conductive components, each formed using fewer than all conductive layers of the multilayered flexible interconnect circuit **100** (e.g., only one conductive layer for each voltage trace). However, having multiple conductive layers allows routing/stacking multiple voltage traces **139** in the same portion of the multilayered flexible interconnect circuit **100**. Furthermore, precise patterning of each conductive layer allows the positioning of multiple voltage traces **139** side-by-side. Voltage traces **139** can be connected to each of the busbars **138** and some form of controller (e.g., a battery management system). More specifically, a voltage trace **139** can be monolithic with one or more busbars **138** or, more specifically, with a portion of the conductive layer that both forms this voltage trace **139** and a portion of the busbar. In some examples, a battery management system is a part of a multilayered flexible interconnect circuit **100**. Alternatively, the busbar portion **102** can be connected to the multilayered flexible interconnect circuit **100** or, more specifically, to the voltage traces **139** of the multilayered flexible interconnect circuit **100** during the fabrication of a battery assembly **180** (e.g., a battery pack).

[0061] Overall, a multilayered flexible interconnect circuit **100** comprises multiple conductive layers stacked along the Z-direction as further described below with reference to FIGS. 1C-1D. Some conductive components (e.g., high-current carrying components such as busbars **138**) may be formed using all or at least two or more conductive layers. Other conductive components (e.g., low-current carrying components such as voltage traces **139**) may be formed using a single conductive layer or at least fewer than all conductive layers.

FIGS. 1C-1D: Examples of Different Portions of Multilayered Flexible Interconnect Circuits

[0062] FIG. 1C is a cross-sectional side view of a multilayered flexible interconnect circuit **100**, in accordance with some examples. The multilayered flexible interconnect circuit **100** comprises a first outer insulator layer **111**, a second outer insulator layer **112**, an inner insulator layer **141**, a first conductive layer **131**, and a second conductive layer **132**. As noted above, the multilayered flexible interconnect circuit **100** comprises at least two conductive layers, e.g., a first conductive layer **131** and a second conductive layer **132**. However, any number of conductive layers are within the scope. In some examples, a multilayered flexible interconnect circuit **100** comprises four, five, six, or even more conductive layers.

[0063] Referring to FIG. 1C, the first outer insulator layer **111**, second outer insulator layer **112**, inner insulator layer **141**, first conductive layer **131**, and second conductive layer **132** collectively

define a busbar portion **102**, a busbar support portion **104**, and a metal-free portion **108** of the multilayered flexible interconnect circuit **100**. In some examples, these components also define an insulated conductor portion **106** of the multilayered flexible interconnect circuit **100**. The structure and function of each of these portions will now be described in more detail. These portions may have different combinations of the first outer insulator layer **111**, second outer insulator layer **112**, inner insulator layer **141**, first conductive layer **131**, and second conductive layer **132** thereby forming different components of the multilayered flexible interconnect circuit **100**. For example, a combination of the busbar portion **102** and busbar support portion **104** forms a busbar **138**. More specifically, the busbar portion **102** represents the exposed portion of the busbar **138** allowing various connections to the busbar. The busbar support portion **104** provides the mechanical support to the busbar portion **102**.

[0064] Referring to FIG. 1C, in the busbar portion **102**, the first conductive layer **131** and the second conductive layer **132** directly interface with each other. In other words, the inner insulator layer **141** is not a part of the busbar portion **102** or, more specifically, the inner insulator layer **141** has an opening that coincides with the busbar portion **102**. The interfacing allows direct interconnection of the first conductive layer **131** and the second conductive layer **132** (e.g., by welding).

[0065] Furthermore, the surface of the first conductive layer **131** facing away from the second conductive layer **132** is exposed, e.g., to form a direct connection to cell terminals **192**. This exposure is provided by an opening in the first outer insulator layer **111**. Similarly, the surface of the second conductive layer **132** facing away from the first conductive layer **131** is exposed, e.g., to allow for welding or other tools to reach the second conductive layer **132** such as while forming a connection to cell terminals **192**. This exposure is provided by an opening in the second outer insulator layer **112**. For example, in a battery assembly, the first conductive layer **131** may be positioned between the second conductive layer **132** and cell terminals **192**. The welding of the first conductive layer **131** to the cell terminals **192** may be performed through the second conductive layer **132** such that this welding also interconnects the first conductive layer **131** and second conductive layer **132**. Stacking and interconnecting the first conductive layer **131** and second conductive layer **132** allows a higher current between the cell terminals **192** connected to these conductive layers. In some examples, the busbar portion **102** provides a current rating of at least 50 Amperes, at least 100 Amperes, or even at least 200 Amperes. In the same or other examples, the collective thickness of all conductive layers forming the busbar portion **102** is at least 200 micrometers, at least 500 micrometers, at least 1 millimeter, or even at least 3 millimeters. This combined thickness depends on the (1) required current ratings, (2) one or more materials of the conductive layers, (3) the thickness of each layer, and (4) the number of layers. It should be noted that separating this thickness into multiple layers allows use different footprints/patterns for each conductive layer thereby monolithically integrating different features into one or more conductive layers.

[0066] To form a busbar portion **102**, an opening is formed in each of the first outer insulator layer **111**, second outer insulator layer **112**, and inner insulator layer **141**. In other words, each of the first outer insulator layer **111**, second outer insulator layer **112**, and inner insulator layer **141** has an opening in the busbar portion **102**. The opening in the inner insulator layer **141** allows the first conductive layer **131** and the second conductive layer **132** to directly interface with each other. The opening in the first outer insulator layer **111** provides access and exposes a portion of the first conductive layer **131**, e.g., to form connections to cell terminals **192**. The opening in the second outer insulator layer **112** provides access and exposes a portion of the second conductive layer **132**, e.g., during welding of the second conductive layer **132** to the first conductive layer **131** and to the cell terminals **192**.

[0067] Referring to FIG. 1C, in some examples, the busbar support portion **104** can partially surround and support the busbar portion **102**. Specifically, before being connected to cell terminal

192, different conductive layers are not interconnected or otherwise attached in the busbar portion **102**. In other words, before being connected to cell terminals **192**, the busbar portion **102** is a stack of multiple conductive layers that directly interface with each other but are not connected in the busbar portion **102**. This connection is provided by the insulator layers in the busbar portion **102**. Specifically, in the busbar support portion **104**, the inner insulator layer **141** is stacked between and directly interfaces the first conductive layer **131** and the second conductive layer **132**, e.g., as shown in FIGS. **1C** and **1D**. More specifically, the inner insulator layer **141** is bonded to and supports both the first conductive layer **131** and the second conductive layer **132**. Furthermore, in the busbar support portion **104**, the first conductive layer **131** is stacked between and directly interfaces the first outer insulator layer **111** and the inner insulator layer **141**. The first conductive layer **131** can be bonded to and supported by both the first outer insulator layer **111** and the inner insulator layer **141**. Finally, in the busbar support portion **104**, the second conductive layer **132** is stacked between and directly interfaces the inner insulator layer **141** and the second outer insulator layer **112**. Again, the second conductive layer **132** can be bonded and supported by both the inner insulator layer **141** and the second outer insulator layer **112**. Overall, all three of the inner insulator layer **141**, the first outer insulator layer **111**, and the second outer insulator layer **112** can be used to support the first conductive layer **131** and second conductive layer **132** relative to each other and other components in the multilayered flexible interconnect circuit **100**, at least in the busbar support portion **104**.

[0068] In some examples, at least one of the first outer insulator layer **111** or the second outer insulator layer **112** or even both the first outer insulator layer **111** and the second outer insulator layer **112** are not present in the busbar support portion **104** thereby exposing one or both of the first conductive layer **131** and second conductive layer **132**. The support to the first conductive layer **131** and the second conductive layer **132** can be provided by at least the inner insulator layer **141**.

[0069] Referring to FIG. **1C**, in some examples, in the metal-free portion **108**, the inner insulator layer **141** is stacked between and directly interfaces the first outer insulator layer **111** and the second outer insulator layer **112**. The metal-free portion **108** may be used for supporting different insulators relative to each other in the multilayered flexible interconnect circuit **100**. Alternatively, the metal-free portion **108** is formed by the first outer insulator layer **111** and the second outer insulator layer **112** but not by the inner insulator layer **141** (i.e., the inner insulator layer **141** may not extend through the metal-free portion **108**). Furthermore, in some examples, the metal-free portion **108** may be formed by the inner insulator layer **141** and only one or the first outer insulator layer **111** and the second outer insulator layer **112**. The metal-free portion **108** can form at least some of the outside edges of the multilayered flexible interconnect circuit **100** (e.g., to position and seal the first conductive layer **131** and second conductive layer **132** away from these edges). In some examples, all conductive layers of the multilayered flexible interconnect circuit **100** are positioned away from the edges of the multilayered flexible interconnect circuit **100**. In this example, the connections to the conductive layers may be formed through various openings in the insulators. Alternatively, at least a portion of the first conductive layer **131** and/or second conductive layer **132** may extend through one or more edges of the multilayered flexible interconnect circuit **100**.

[0070] Referring to FIG. **1C**, in some examples, the multilayered flexible interconnect circuit **100** comprises an insulated conductor portion **106**. Unlike the busbar support portion **104**, which comprises multiple conductive layers (e.g., all conductive layers forming the adjacent the busbar portion **102**), an insulated conductor portion **106** can include fewer than all conductive layers, such as only one conductive layer, e.g., a second conductive layer **132** as shown in FIG. **1C**. Specifically, in the insulated conductor portion **106**, the second conductive layer **132** is stacked between and directly interfaces the inner insulator layer **141** and the second outer insulator layer **112**. The inner insulator layer **141** is stacked between and directly interfaces the second conductive layer **132** and the first outer insulator layer **111**. Referring to FIG. **1C**, in some examples, the insulated conductor

portion **106** is positioned between the busbar support portion **104** and metal-free portion **108**. In the same or other examples, the busbar support portion **104** is positioned between the busbar portion **102** and the metal-free portion **108**.

[0071] In some examples, each of the first outer insulator layer **111**, the second outer insulator layer **112**, the inner insulator layer **141**, the first conductive layer **131**, and the second conductive layer **132** is formed from the same starting sheet. As such, each of the first outer insulator layer **111**, the second outer insulator layer **112**, the inner insulator layer **141**, the first conductive layer **131**, and the second conductive layer **132** may have the same thickness and composition throughout the entire footprint of the multilayered flexible interconnect circuit **100**. In fact, some layers (e.g., the first outer insulator layer **111** and/or the second outer insulator layer **112**) may be monolithic throughout the entire footprint of the multilayered flexible interconnect circuit **100**. Other layers may be cut into disjointed components, e.g., busbars, insulating patches, etc. It should be noted that each layer is individually patterned to form different portions/components of the multilayered flexible interconnect circuit **100**.

[0072] Referring to FIG. 1D, in some examples, each of the first outer insulator layer **111** and the second outer insulator layer **112** comprises a polymer base and an adhesive layer covering a surface of and supported by the polymer base. The polymer base comprises one or more polymer selected from the group consisting of polyimide (PI), polyethylene naphthalate (PEN), polyethylene terephthalate (PET), polymethyl methacrylate (PMMA), ethyl vinyl acetate (EVA), polyethylene (PE), polyvinyl fluoride (PVF), polyamide (PA), and/or polyvinyl butyral (PVB). The adhesive layer comprises one or more of epoxy and polyurethane. In more specific examples, in the busbar support portion **104**, the adhesive layer of the first outer insulator layer **111** directly interfaces and is adhered to the first conductive layer **131**. The adhesive layer of the second outer insulator layer **112** directly interfaces and is adhered to the second conductive layer **132**.

[0073] Referring to FIG. 1D, in some examples, the inner insulator layer **141** comprises an inner polymer base, a first inner adhesive layer, and a second inner adhesive layer. The inner polymer base is positioned between and supports each of the first inner adhesive layer and the second inner adhesive layer. The inner polymer base comprises one or more polymer selected from the group consisting of polyimide (PI), polyethylene naphthalate (PEN), polyethylene terephthalate (PET), polymethyl methacrylate (PMMA), ethyl vinyl acetate (EVA), polyethylene (PE), polyvinyl fluoride (PVF), polyamide (PA), and/or polyvinyl butyral (PVB). Each of the first inner adhesive layer and the second inner adhesive layer comprises one or more of epoxy and polyurethane. In more specific examples, in the busbar support portion **104**, the first inner adhesive layer of the inner insulator layer **141** directly interfaces and is adhered to the first conductive layer **131**. Furthermore, the adhesive layer of the second outer insulator layer **112** directly interfaces and is adhered to the second conductive layer **132**.

[0074] In some examples, each of the first conductive layer **131** and the second conductive layer **132** comprises aluminum. However, other metals (e.g., copper) are also within the scope. In some examples, all conductive layers (e.g., both the first conductive layer **131** and the second conductive layer **132**) are formed from the same material, e.g., aluminum, copper, or the like. Alternatively, different metals may be used for the first conductive layer **131** and the second conductive layer **132**. In general, the first conductive layer **131** and the second conductive layer **132** may be formed from any conductive material that is sufficiently conductive (e.g., a conductivity being greater than 10^6 S/m or even greater than 10^7 S/m to allow for current flow through the foil with low power loss).

[0075] In some examples, each of the first conductive layer **131** and the second conductive layer **132** has a thickness of 100-400 micrometers or, more specifically, 200-300 micrometers. The total thickness of all conductive layers forming a busbar can be at least 400 micrometers, at least 600 micrometers, at least 800 micrometers, or even at least 1,000 micrometers such as 500-2,000 micrometers or, more specifically, 600-1,200 micrometers.

[0076] This thickness depends on the current-carrying requirements from the busbar. In some examples, all conductive layers in a multilayered flexible interconnect circuit **100** have the same thickness, e.g., as shown in FIG. **1D**. Alternatively, at least one conductive layer is thicker than another conductive layer in the multilayered flexible interconnect circuit **100**.

[0077] Referring to FIG. **1D**, in some examples, a multilayered flexible interconnect circuit **100** further comprises a circuit bonding layer **113**, e.g., a double-sides pressure sensitive adhesive (PSA) layer.

FIGS. **2A-3B**: Examples of Multilayered Flexible Interconnect Circuits Comprising Interconnecting Units

[0078] The functional and structural aspects of multilayered flexible interconnect circuits comprising interconnecting units will now be described in the context of FIGS. **2A-2F**. Specifically, FIG. **2A** is an expanded top view of a portion of a multilayered flexible interconnect circuit comprising a plurality of busbars **120** and connected components, in accordance with some examples. The multilayered flexible interconnect circuit **100** comprises a plurality of busbars **120**, a plurality of interconnecting-unit control lines **165**, and a plurality of interconnecting units **160**. Each one of the plurality of busbars **120** is defined by a contact portion **125**, a first tab portion **126**, and a second tab portion **127**. The plurality of interconnecting-unit control lines **165** comprises more than one interconnecting-unit control line **161**. Each one of the plurality of interconnecting units **160** is connected to two of the plurality of busbars **120** and is separately connected to an interconnecting-unit control line **161**. Each one of the plurality of interconnecting units **160** is stacked between and connected to the first tab portion **126** of one of the plurality of busbars **120** and the second tab portion **127** of another one of the plurality of busbars **120**. In some examples, at least one of the plurality of interconnecting units **160** is stacked between either a first tab portion **126** or a second tab portion **127** of one of the plurality of busbars **120** and the other of either a first tab portion **126** and a second tab portion **127** of the another one of the plurality of busbars **120**. In some further examples one of the plurality of busbars **120** is a first busbar **121** and the other one of the plurality of busbars **120** is a second busbar **122**.

[0079] Each one of the plurality of interconnecting units **160** electrically disconnects the first tab portion **126** first tab portion **126** and the second tab portion **127** before receiving an electrical input through the interconnecting-unit control line **161**. In other words, when any one of the plurality of interconnecting units **160** does not receive an electrical input via the interconnecting-unit control line **161**, the first tab portion **126** and the second tab portion **127** are electrically disconnected. When one of the plurality of interconnecting units **160** receives an electrical input through the interconnecting-unit control line **161**, the plurality of interconnecting units **160** electrically connects the first tab portion **126** and the second tab portion **127** it is connected with. Each one of the plurality of interconnecting units **160** is configured to electrically connect the two of the plurality of busbars **120** it is connected to after receiving the electrical input through the interconnecting-unit control line **161** it is connected to.

[0080] In some examples, the plurality of busbars **120** comprises a first busbar **121** and a second busbar **122**. Each one of the plurality of busbars **120** comprises a first conductive layer **131** and a second conductive layer **132**. In some examples, the first tab portion **126** is formed from the first conductive layer **131** and the second tab portion **127** is formed from the second conductive layer **132**. In these examples, this may provide manufacturing benefits. For example, it may simplify manufacturing of the multilayered flexible interconnect circuit **100** if the plurality of interconnecting units **160** can be placed between conductive layers used to form the first conductive layer **131** and second conductive layer **132**.

[0081] As described above, each one of the plurality of interconnecting units **160** is physically connected to two of the plurality of busbars **120** but maintains the two of the plurality of busbars **120** in electrical disconnection before receiving an electrical input through the interconnecting-unit control line **161**. Shown in FIG. **3A** is a schematic cross-sectional side view of one of the plurality

of interconnecting units **160** connecting a first busbar **121** and a second busbar **122**. Shown are a first conductive layer **131** and a second conductive layer **132** from a first busbar **121** and a first conductive layer **131** and a second conductive layer **132** from a second busbar **122**. Also shown is the interconnecting-unit control line **161** connected to the one of the plurality of interconnecting units **160**. The first busbar **121** is connected to a terminal of a battery cell and the second busbar **122** is connected to another terminal of the same battery cell. In FIG. 3A, no electrical input is received via the interconnecting-unit control line **161**. The one of the plurality of interconnecting units **160** is configured to maintain electrical disconnection between the first busbar **121** and the second busbar **122**. Electrical current flows from the first busbar **121** into and through the battery cell and into the second busbar **122**.

[0082] The first busbar electrically connects cell terminal **192** (not shown) of a battery cell (not shown) to cell terminal **191** of another battery cell. Second busbar **122** electrically connects cell terminal **192** to cell terminal **191** of yet another battery cell. The plurality of busbars **120** connects the adjacent battery cells shown in FIG. 2A in series. Other arrangements in which some of the plurality of busbars **120** connect battery cells or series sets of battery cells in parallel are also within the scope. Shown in FIGS. 2B and 2C are expanded top view (2B) and expanded side view (2C) of second busbar **122**. As shown for second busbar **122**, The contact portion **125** is formed by both the first conductive layer **131** and the second conductive layer **132**, stacked in the contact portion **125**. The first tab portion **126** is formed by the first conductive layer **131** such that the second conductive layer **132** does not extend to the first tab portion **126**. The second tab portion **127** is formed by the second conductive layer **132** such that the first conductive layer **131** does not extend to the second tab portion **127**.

[0083] FIG. 2G is an expanded top view of a portion of a multilayered flexible interconnect circuit comprising a plurality of busbars **120** and connected components, in accordance with some examples. In FIG. 2A, the plurality of interconnecting units **160** are placed between the first conductive layer **131** and second conductive layer **132** of individuals of the plurality of busbars **120** attached to separate terminals of the same battery cells **190**. In FIG. 2G, the plurality of interconnecting units **160** are placed between the first conductive layer **131** and second conductive layer **132** of busbars that are attached to terminals of separate battery cells **190**. In this configuration, electrically connecting the first conductive layer **131** and the second conductive layer **132** allows current to flow around a disabled battery cell and an adjacent battery cell. As shown in FIG. 2G, in these examples, a portion of the first conductive layer **131** and a portion of the second conductive layer **132** may extend to form a path, with one of the plurality of interconnecting units **160**, around one or more other busbars.

[0084] Where the configuration of interconnecting units **160** and plurality of busbars **120** shown in FIG. 2A will form an electrical short pathway between the two terminals of a battery cell **190**, the configuration in shown in FIG. 2G will not form an electrical short pathway between the terminals of a battery cell **190**. This may provide a benefit in situations where a battery cell **190** has failed and is charged to a high state of charge. In such a situation, creating a short pathway between the two terminals of a battery cell **190** may be desirable if the battery cell **190** has failed in a mode forming an open circuit between the terminals of the two battery cells **190**. However, creating a short pathway between the two terminals of a battery cell **190** may be dangerous if the battery cell **190** has failed in another way. A battery cell **190** may rapidly discharge its stored electrical energy through the electrical connection formed between the first conductive layer **131** and the second conductive layer **132**. This may create a dangerous temperature increase due to resistive heating of the electrical short pathway. Instead, passing the electrical current from the rest of the interconnected battery cells **190** around the damaged battery cell and an adjacent battery cell **190** may be desirable. Importantly, a multilayered flexible interconnect circuit **100** may comprise both the configuration shown in FIG. 2A and the configuration shown in

[0085] FIG. 2G. This may be a benefit provided by the construction the multilayered flexible

interconnect circuit **100**. For example, if the plurality of busbars **120** are formed from four conductive layers, two layers may be used to form a first conductive layer **131** and a second conductive layer **132** configured to connect a first busbar **121** and a second busbar **122** that are adjacent. Two other layers may be used to form a first conductive layer **131** and a second conductive layer **132** configured to connect a first busbar **121** and a second busbar **122** that are not adjacent.

[0086] Each of the plurality of interconnecting units **160** is configured to electrically connect the two of the plurality of busbars **120** after receiving the electrical input. FIG. 3B is a schematic view of one of the plurality of interconnecting units **160** connecting a first busbar **121** and a second busbar **122**, after an electrical input is received via the interconnecting-unit control line **161**. In this example, the one of the plurality of interconnecting units **160** is configured to electrically connect the first busbar **121** and second busbar **122**. Electrical current flows from the first busbar **121** to the second busbar **122** and does not flow through the battery cell. Details of configurations of the plurality of interconnecting units **160** will be described in more detail below. In some examples, the electrical input has a magnitude of 0.25-12 Volts, 0.5-9 Volts, or even 1-5 Volts.

[0087] In some examples, the first conductive layer **131** and second conductive layer **132** are directly interfacing each other in the contact portion **125**, as shown in FIG. 2D. In other examples, the first conductive layer **131** and second conductive layer **132** are separated from each other in the contact portion **125** by an insulation layer **135**, as shown in FIG. 2E.

[0088] In some examples, the interconnecting-unit control line **161** is a dedicated trace, separate from any other conductor traces, for example, a voltage trace **139**. In other examples, the interconnecting-unit control line **161** is a voltage trace **139**. In some examples, one interconnecting-unit control line **161** is connected to more than one of the plurality of interconnecting units **160**. In these examples, an electrical signal transmitted by the interconnecting-unit control line **161** can control more than one of the plurality of interconnecting units **160** at once. In other examples, at least one of the plurality of interconnecting units **160** is connected to more than one interconnecting-unit control line **161**. In these examples, electrical signals may be transmitted separately by more than one interconnecting-unit control line **161** to one of the plurality of interconnecting units **160**.

[0089] In some examples, the plurality of busbars **120** may be positioned such that they are suspended above the surface of the battery cells **190** where the cell terminals **191** and cell terminals **192** are positioned. The insulator layers of the multilayered flexible interconnect circuit **100** may have a vaporization point lower than temperatures expected to exist near the surface of a battery cell **190** undergoing thermal run-away. In these examples, the plurality of interconnecting units **160** is located either close to the surface of the battery cell **190** or directly over a cell vent. A cell vent is a pressure release valve built into the container of a battery cell **190** to release hot metal and gases in the event of a thermal run-away. The high temperatures of either the surface of the battery cell **190** or escaping from a cell vent will melt and vaporize the insulator layers, allowing the first conductive layer **131** and second conductive layer **132** to make electrical contact. This can provide protection to the battery assembly **180** by removing a damaged battery cell **190** from electrical connection with the battery assembly **180**. This may provide protection before an electrical input is received by the plurality of interconnecting units **160**.

FIGS. 4A-4C: Examples of Interconnecting Units

[0090] Specific examples of interconnecting units will now be described in detail with reference to FIGS. 4A-4D. FIG. 4A presents a schematic block diagram illustrating the relationships of components general to the interconnecting unit examples. Each one of the plurality of interconnecting units **160** comprises an interconnecting-unit control element **164** that is electrically connected to an interconnecting-unit first contact **162**, an interconnecting-unit second contact **163**, and an interconnecting-unit control line **161**. As described above, each one of the plurality of interconnecting units **160** is electrically connected to both a first conductive layer **131** and second

conductive layer **132** of the plurality of busbars **120**. As shown in FIG. 4A, more specifically, the interconnecting-unit first contact **162** is electrically connected to a first conductive layer **131** and the interconnecting-unit second contact **163** is electrically connected to a second conductive layer **132**. Different electrical attachments are within the scope, including but not limited to a weld, a conductive pressure-sensitive adhesive, and a conductive curable adhesive. The interconnecting-unit control element **164** is configured to provide electrical insulation between the interconnecting-unit first contact **162** and interconnecting-unit second contact **163** until an electrical input is received via the interconnecting-unit control line **161**. The interconnecting-unit control element **164** is configured to electrically connect the interconnecting-unit first contact **162** and interconnecting-unit second contact **163** after receiving an electrical input via the interconnecting-unit control line **161**. When the interconnecting-unit first contact **162** and interconnecting-unit second contact **163** are electrically connected, the first conductive layer **131** and second conductive layer **132** will be in electrical communication.

[0091] In some examples, each of the plurality of interconnecting units **160** reversibly electrically connect the first conductive layer **131** and second conductive layer **132**. In other words, at least one of the plurality of interconnecting units **160** is configured to electrically connect the two of the plurality of busbars **120** it is connected to after receiving the electrical input through the interconnecting-unit control line **161** and configured to electrically disconnect the two of the plurality of busbars **120** again when the electrical input ceases. A multilayered flexible interconnect circuit **100** comprising reversibly connecting interconnecting-unit control elements **164** may provide advantages in operation. For example, if more battery cells fail after one interconnecting-unit control element **164** is configured to connect two of the plurality of busbars **120**, the interconnecting-unit control element **164** may be re-configured if needed to change the routing of electrical current when other interconnecting-unit control elements **164** are configured to connect busbars.

[0092] An example of a reversibly connecting interconnecting-unit control element **164** is shown schematically in FIG. 4B. In FIG. 4B, the interconnecting-unit control element **164** is a transistor **305**. In this example, the transistor **305** comprises three electrical connections: a source **310**, a drain **315**, and a gate **320**. In FIG. 4B, the source **310** is electrically connected to the interconnecting-unit first contact **162** and the drain **315** is electrically connected to the interconnecting-unit second contact **163**. In other examples, the source **310** is electrically connected to the interconnecting-unit second contact **163** and the drain **315** is electrically connected to the interconnecting-unit first contact **162**. When no electrical input greater than the turn-on voltage for the transistor **305** is received at the gate **320** via the interconnecting-unit control line **161**, the transistor **305** provides a high-impedance resistance to electric current flow between the source **310** and the drain **315**. When an electrical input greater than the turn-on voltage for the transistor **305** is received at the gate **320**, the transistor **305** allows current to pass from the source **310** to the drain **315**. This configures the one of the plurality of interconnecting units **160** to electrically connect the first conductive layer **131** and second conductive layer **132**. In some examples, the electrical input has a voltage of 0.5-12 V, 1-9 V, or even 3-5 V. In some examples, the power passing from the interconnecting-unit first contact **162** to the interconnecting-unit second contact **163** through the transistor **305** is 0.2-8 kilowatts (kW), 0.3-5 kW, or even 1-4 kW.

[0093] In some examples, the transistor **305** is an external electrical component attached to the multilayered flexible interconnect circuit **100** by soldering, welding, or a conductive adhesive. In these examples, the construction of the multilayered flexible interconnect circuit **100** may provide benefits of simplified manufacturing. The conductor layers to which the source **310**, drain **315**, and gate **320** are connected to may be designed with thickness optimized for making soldered joints. In other examples, the transistor **305** is fabricated within one of the layers of the multilayered flexible interconnect circuit **100**.

[0094] In some examples, each of the plurality of interconnecting units **160** is a normally-open

relay **340**. In these examples, this may provide benefits of lower manufacturing cost and design flexibility. Normally-open relays may be lower in cost than transistors designed for the same maximum electrical current service. In addition, a larger number of suitable normally-open relays may be available than transistors, which may provide improved design flexibility. However, normally-open relays may have larger packages than transistors, requiring greater space on the multilayered flexible interconnect circuit **100**. The normally-open relay **340** comprises a first pole **345**, a second pole **350**, and an actuation lead **355**, as shown schematically in FIG. 4D. The first pole **345** and the second pole **350** are electrically disconnected when no electrical input is applied to the actuation lead **355**. The first pole **345** and the actuation lead **355** may be electrically connected when some electrical inputs are applied to the actuation lead **355**. The properties of the electrical inputs required to connect the first pole **345** and the second pole **350** depend on the design of the normally-open relay **340**. The first pole **345** is electrically interconnected with either of the first tab portion **126** of the first busbar **121** and the second tab portion **127** of the second busbar **122**. The second pole **350** is electrically interconnected with the other of either the first tab portion **126** of the first busbar **121** and the second tab portion **127** of the second busbar **122**. The actuation lead **355** is electrically interconnected with the interconnecting-unit control line **161**. Each normally-open relay **340** is configured to reversibly electrically connect the first busbar **121** and the second busbar **122** after receiving the electrical input through the actuation lead **355**. In some examples, the normally-open relay **340** is an external electrical component attached to the multilayered flexible interconnect circuit **100** by soldering, welding, or a conductive adhesive.

[0095] In some examples, each one of the plurality of interconnecting units **160** comprises one or more voids **136** and an external pressure component **170**, as shown in FIG. 2F. The one or more voids **136** creates an open volume from the interconnecting-unit first contact **162** to the interconnecting-unit second contact **163**. The external pressure component **170** is electrically connected to the interconnecting-unit control line **161**. When no electrical input is received by the external pressure component **170**, the one of the plurality of interconnecting units **160** physically separates the interconnecting-unit first contact **162** and the interconnecting-unit second contact **163**. The external pressure component **170** is configured to apply pressure to press the interconnecting-unit first contact **162** and the interconnecting-unit second contact **163** towards one another. When this pressure is applied, the one of the plurality of interconnecting units **160** is compressed and the interconnecting-unit first contact **162** and the interconnecting-unit second contact **163** make physical and electrical contact at the one or more voids **136**. The thickness and compression resistance of the material of the one of the plurality of interconnecting units **160** and the number, diameter, and arrangement of the one or more voids **136** can be chosen to minimize contact of the interconnecting-unit first contact **162** to the interconnecting-unit second contact **163** when no pressure is applied. These same qualities can also be chosen to maximize the likelihood and quality of the electrical connection made when pressure is applied. For example, an arrangement of a larger number of smaller diameter voids may better prevent unwanted electrical contact than one larger diameter void. The thickness of the one of the plurality of interconnecting units **160** in the direction separating the interconnecting-unit first contact **162** and the interconnecting-unit second contact **163** may be chosen to be sufficiently thick to prevent sparking between the contacts at the highest expected voltage difference between the two.

[0096] In some examples, each of the plurality of interconnecting units **160** is configured to irreversibly electrically connect the first conductive layer **131** and second conductive layer **132**. A multilayered flexible interconnect circuit **100** comprising irreversibly connecting interconnecting-unit control elements **164** may provide advantages in operation. For example, continuous electrical current must be applied to maintain an electrical input to keep a reversibly connecting interconnecting-unit control element **164**. However, once an irreversibly connecting interconnecting-unit control element **164** connects a first conductive layer **131** and a second conductive layer **132**, no further electrical current is required to maintain the connection. In this

way, more stored electrical power in the battery assembly **180** or an auxiliary power source can be preserved. This may be especially desirable in operation of a limp-home mode for an electric vehicle in a situation where many battery cells **190** have been damaged.

[0097] In some examples, each one of the plurality of interconnecting units **160** is a heat-actuated interconnector **370**. In these examples, each heat-actuated interconnector **370** comprises a first connecting portion **375**, a second connecting portion **380**, a meltable portion **385**, and a heating element **390**. The first connecting portion **375** is electrically interconnected with either the first tab portion **126** of the first busbar **121** and the second tab portion **127** of the second busbar **122**. The second connecting portion **380** is electrically interconnected with the other of either the first tab portion **126** of the first busbar **121** and the second tab portion **127** of the second busbar **122**. The meltable portion **385** is positioned between the first connecting portion **375** and the second connecting portion **380** as shown in FIG. 2E. In some examples, the heating element **390** is positioned within the meltable portion **385**. In some other examples, the heating element **390** is positioned between the meltable portion **385** and either the first connecting portion **375** or the second connecting portion **380**. In some other examples, the heating element **390** is positioned either on the opposite side of the first connecting portion **375** from the second connecting portion **380** or on the opposite side of the second connecting portion **380** from the first connecting portion **375**. This is also shown in FIG. 2E. The heating element **390** is electrically interconnected with the interconnecting-unit control line **161**. The heating element **390** is configured to generate heat and melt the meltable portion **385** after receiving the electrical input through the interconnecting-unit control line **161**. When the meltable portion **385** is melted, the first tab portion **126** and second tab portion **127** are irreversibly electrically connected, thereby connecting the first busbar **121** and the second busbar **122**. In some examples, external pressure may be applied by an external component to press the first tab portion **126** and the second tab portion **127** together when the meltable portion **385** is melted. The meltable portion **385** may comprise, for example, a heat-activated solder paste.

[0098] In some examples, the heating element **390** is either the first connecting portion **375** or the second connecting portion **380**. In these examples, the heating element **390** may be patterned to form, for example, a resistive heater. In these examples, when the heating element **390** receives the electrical input, the heating element **390** generates sufficient heat to melt the meltable portion **385**. In some examples, in which insulation layer **135** separates the conductor layer of the heating element **390** and the meltable portion **385**, the heating element **390** generates sufficient heat to melt the insulation layer **135** positioned between the heating element **390** and the meltable portion **385**.

[0099] In some examples, the meltable portion **385** may comprise a conductive adhesive. The conductive adhesive may cure when heat is applied by the heating element **390**, activating the adhesive and rendering it electrically conductive. In some other examples, the conductive adhesive may cure when an electrical input is received through the interconnecting-unit control line **161**, rendering it electrically conductive. The conductive adhesive may be, for example, an anisotropic conductive adhesive such as 124-19C119-44, available from Creative Materials, Ayer, MA.

FIG. 5: Examples of Battery Assemblies

[0100] FIG. 5 is a schematic block diagram of a battery assembly **510** comprising a plurality of interconnecting units **160** in accordance with some examples. Battery assembly **510** comprises a battery management system **520**, a battery management system power source **530**, a multilayered flexible interconnect circuit **100**, and a plurality of battery cells **540**. The battery management system **520** is electrically connected to the battery management system power source **530**. The battery management system power source **530** provides electrical power that the battery management system **520** uses to operate. The multilayered flexible interconnect circuit **100** comprises a plurality of busbars **120** and a plurality of interconnecting units **160**. Each of the plurality of busbars **120** comprises a first conductive layer **131** and a second conductive layer **132**. Each of the plurality of busbars **120** is defined by a contact portion **125**, a first tab portion **126**, and a second tab portion **127**. The contact portion **125** is formed by both the first conductive layer **131**

and the second conductive layer **132**, stacked in the contact portion **125**. The first tab portion **126** is formed by the first conductive layer **131** such that the second conductive layer **132** does not extend to the first tab portion **126**. The second tab portion **127** is formed by the second conductive layer **132** such that the first conductive layer **131** does not extend to the second tab portion **127**. The plurality of interconnecting units **160** comprises an interconnecting unit stacked between and connected to each of the first tab portion **126** of the first busbar **121** and the second tab portion **127** of the second busbar **122**. Examples of the components and configurations of the plurality of interconnecting units **160** are described above. Each one of the plurality of battery cells **540** comprises two cell terminals **192**. Each one of the plurality of busbars **120** is connected to one terminal of each of two of the plurality of battery cells **540**. Each interconnecting-unit control line **161** is electrically connected to the battery management system **520**. In some examples, the connection between each interconnecting-unit control line **161** and the battery management system **520** is a direct electrical connection. In other examples, the connection is made through an intervening electronic bus that allows selectable addressing of each interconnecting-unit control line **161**. Each one of the plurality of interconnecting units **160** maintains the connected first busbar **121** and second busbar **122** that it is connected to electrically disconnected from each other before receiving an electrical input through the interconnecting-unit control line **161**. Each one of the plurality of interconnecting units **160** is configured to electrically connect the first busbar **121** and the second busbar **122** it is connected to after receiving the electrical input through the interconnecting-unit control line **161**.

[0101] In some examples, the multilayered flexible interconnect circuit **100** further comprises a plurality of voltage traces **139**. In these examples, each one of the plurality of voltage traces **139** electrically connects one of the plurality of busbars **120** to the battery management system **520**.

[0102] In some examples, the battery management system power source **530** is an auxiliary battery assembly that is separate from the plurality of battery cells **540**. In some examples, one or more battery cells **190** of the plurality of battery cells **540** are the battery management system power source **530**. For example, two battery cells **190** from the plurality of battery cells **540**, connected in parallel, may be the battery management system power source **530**. This example may provide benefits in saving cost, space, and weight by avoiding inclusion of an auxiliary battery assembly in an electric vehicle. In these examples, if one of the two battery cells **190** fails, the other of the two battery cells **190** will still provide sufficient electrical power for the battery management system **520** to operate.

FIG. 6: Examples of Methods of Fabricating Multilayered Flexible Interconnect Circuits

[0103] FIG. 6 is a process flowchart corresponding to method **600** of fabricating a multilayered flexible interconnect circuit **100**, in accordance with some examples. Various examples and features of multilayered flexible interconnect circuits **100** are described above. Specifically, a multilayered flexible interconnect circuit **100** comprises at least two conductive layers, e.g., a first conductive layer **131** and a second conductive layer **132**. These conductive layers can have different patterns to provide various features described above. Furthermore, the multilayered flexible interconnect circuit **100** also comprises a first outer insulator layer **111**, a second outer insulator layer **112**, and an inner insulator layer **141** that is used to support and, in some parts, to insulate the first conductive layer **131** and the second conductive layer **132**.

[0104] Method **600** may comprise (block **610**) patterning each conductive layer and, separately, (block **620**) patterning each insulator layer, and (block **625**) patterning each interconnect unit. In fact, each layer of a multilayered flexible interconnect circuit **100** can be patterned individually prior to laminating these layers together. In some examples, various temporary substrates may be used for these patterning operations. A temporary substrate can be used to support the patterned layer during and after the patterning operation. The temporary substrate is removed when the patterned layers are stacked.

[0105] As such, in some examples, method **600** or, more specifically, (block **610**) patterning each

conductive layer comprises (block **612**) laminating a first metal sheet to a first temporary substrate and (block **614**) patterning the first metal sheet, while the first metal sheet remains laminated on the first temporary substrate thereby forming a first conductive layer **131**. For example, initially, the metal sheet may be a continuous self-supporting metal foil that can be processed and handled without any additional support. The pattern of the first conductive layer **131** may include disjointed components (e.g., busbar portions), narrow conductive traces, and/or other features that can be self-supported. Unlike the first conductive layer **131**, the temporary substrate is not patterned (e.g., the temporary substrate may remain as a continuous sheet). In some examples, the same temporary substrate may support multiple instances of first conductive layers **131** (e.g., used for the production of multiple units of multilayered flexible interconnect circuits **100**). Various patterning techniques are within the scope. For example, conductive layers can be patterned using chemical etching, mechanical cutting, laser cutting, and the like.

[0106] Similarly, method **600** or, more specifically, (block **610**) patterning each conductive layer comprises (block **616**) laminating a second metal sheet to a second temporary substrate and (block **618**) patterning the second metal sheet to a second temporary substrate thereby forming a second conductive layer **132**. It should be noted that the pattern of the first conductive layer **131** is different from the pattern of the second conductive layer **132**. For example, a busbar (formed by both conductive layers) may have only one conductive layer protruding away and forming a voltage trace.

[0107] Method **600** may also comprise (block **620**) patterning one or more insulator layers to form various openings. These openings (in the outer insulator layers) are used to provide access to the conductive layers at some locations (e.g., busbars). Furthermore, the openings in inner insulator layers allow the interconnection of the conductive layers at some locations (e.g., busbars). In some examples, after patterning insulator layers, at least one insulator represents a continuous structure (that is able to self-support and does not require any additional structures).

[0108] Method **600** may also comprise (block **625**) patterning one or more interconnect layers. In some examples, these layers are conductive layers that provide leads or pads for later attachment of interconnect devices (e.g. Transistor **305**). In some examples, these layers are polymer or adhesive layers patterned to form components of the plurality of interconnecting units **160** (e.g. Melttable portion **385**).

[0109] Method **600** further comprises (block **630**) stacking and laminating the first conductive layer **131**, second conductive layer **132**, first outer insulator layer **111**, second outer insulator layer **112**, and inner insulator layer **141** thereby forming a multilayered flexible interconnect circuit **100**. In some examples, this stacking and laminating operation comprises (block **634**) laminating the first conductive layer **131** to a first outer insulator layer **111** and/or inner insulator layer **141** and removing any temporary support from the first conductive layer **131**. After laminating to one or both of the insulator layers, the first conductive layer **131** is supported by one or both of the insulator layers and the temporary support is no longer needed. Similarly, the second conductive layer **132** may be laminated to a second outer insulator layer **112** and/or inner insulator layer **141** and remove any temporary support from the second conductive layer **132**. A combination of a patterned conductive layer (e.g., a first conductive layer **131**, a second conductive layer **132**) and one or two insulators laminated to this patterned conductive layer may be referred to as a metal-insulator unit. Forming these insulator units prior to forming a full stack of the multilayered flexible interconnect circuit **100** allows the removal of any temporary substrates. Method **600** then proceeds with (block **637**), stacking interconnect units. Method **600** then proceeds with (block **638**) stacking and laminating these metal-insulator units, each comprising at least one patterned conductive layer. In some examples, a patterned conductive layer comprises two patterned conductive layers, e.g., an inner insulator layer **141** with a first conductive layer **131** laminated to one side and a second conductive layer **132** laminated to the other side.

[0110] In some examples, method **600** comprises (block **640**) attaching a support unit, e.g., to a

first outer insulator layer **111**. For example, a support unit may have an adhesive layer interfacing the first outer insulator layer **111** during this operation.

[0111] In some examples, method **600** comprises (block **650**) attaching external components. These components are components of the plurality of interconnecting units **160**. For example, the transistor **305** or the normally-open relay **340** may be soldered or welded to the multilayered flexible interconnect circuit **100**.

FIG. 7: Examples of Methods of Operating Battery Packs Comprising Multilayered Flexible Interconnect Circuit with a Plurality of Interconnecting Units

[0112] FIG. 7 is a process flow chart corresponding to method **700** of operating a battery assembly **510**. Various examples and features of battery assembly **510** have been described above. Method **700** may commence with (block **710**) providing a battery pack comprising a plurality of busbars, a plurality of interconnecting units, a plurality of battery cells comprising cell terminals, a plurality of voltage traces, and a battery management system. The battery cell **190** is one component of a battery assembly **510**, as shown in FIG. 5. The battery assembly comprises a plurality of busbars **120**, a plurality of interconnecting units **160**, a plurality of battery cells **540**, a plurality of voltage traces **139**, and a battery management system **520**. The plurality of busbars **120** comprises a first busbar **121** and a second busbar **122**. Each of the plurality of interconnecting units **160** is connected to two of the plurality of busbars **120** and separately connected to an interconnecting-unit control line **161**. Each one of the plurality of battery cells **540** comprises cell terminals **192**. Each of the plurality of busbars **120** comprises a first conductive layer **131** and a second conductive layer **132**, and is defined by a contact portion **125**, a first tab portion **126**, and a second tab portion **127**. The contact portion **125** is formed by both the first conductive layer **131** and the second conductive layer **132**, stacked in the contact portion **125**. The first tab portion **126** is formed by the first conductive layer **131** such that the second conductive layer **132** does not extend to the first tab portion **126**. The second tab portion **127** is formed by the second conductive layer **132** such that the first conductive layer **131** does not extend to the second tab portion **127**. The cell terminals **192** of each one of the plurality of battery cells **540** are each electrically connected to different ones of the plurality of busbars **120**. Each of the interconnecting-unit control lines **161** is electronically connected with the battery management system **520**. Each of the voltage traces **139** electrically connects one of the plurality of busbars **120** to the battery management system **520**. The plurality of interconnecting units **160** comprise an interconnecting unit **160** stacked between and connected to each of the first tab portion **126** of the first busbar **121** and the second tab portion **127** of the second busbar **122**. Each one of the plurality of interconnecting units **160** maintains the two of the plurality of busbars **120** electrically disconnected from each other before receiving an electrical input through the interconnecting-unit control line **161**. Each one of the plurality of interconnecting units **160** is configured to electrically connect the two of the plurality of busbars **120** after receiving the electrical input through the interconnecting-unit control line **161**. Method **700** may then proceed with (block **720**), determining at the battery management system if the state of health (SOH) of any one of the plurality of battery cells is unsatisfactory. The battery management system **520** is configured to determine if the SOH of any of the plurality of battery cells **540** is unsatisfactory by measuring voltages at each of the plurality of busbars **120**. In some examples, the battery management system **520** is configured to determine battery cell SOH by comparing voltages measured at each of the plurality of busbars **120** to a predetermined list of voltages.

[0113] In some examples, the method may proceed with (block **730**) determining at the battery management system **520** the state of charge (SOC) of any one of the plurality of battery cells **540** determined by the battery management system **520** to have unsatisfactory SOH.

[0114] Method **700** may proceed by (block **740**) selecting at the battery management system **520** which of the plurality of interconnecting units **160** to provide an electrical input to. The features and benefits of activating individual ones of the plurality of interconnecting units **160** have been described above. In some examples, the battery management system **520** may select one or more of

the plurality of interconnecting units **160** by comparing the identification of battery cells **190** that have unsatisfactory SOH with a predetermined table that indicates which of the plurality of interconnecting units **160** to activate. If SOC has also been determined by the battery management system **520**, this information may also be applied to the selection of which of the plurality of interconnecting units **160** to provide an electrical input to. As described above, it may be desirable to remove a battery cell **190** from electrical connection with the battery assembly **510** by activating one interconnecting unit **160** if the SOC is high and a different interconnecting unit **160** if the SOC is low.

[0115] Method **700** may proceed by (block **750**) providing, by the battery management system **520**, an electrical input to the selected ones of the plurality of interconnecting units **160** via at least one of the interconnecting-unit control line **161**.

Conclusion

[0116] It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described examples (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings presented herein. Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of some examples and are by no means limiting and are merely examples. Many examples and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description.

Claims

1. A multilayered flexible interconnect circuit comprising: a plurality of busbars, comprising a first busbar and a second busbar, wherein each one of the plurality of busbars is defined by a contact portion, a first tab portion, and a second tab portion; a plurality of interconnecting-unit control lines, comprising an interconnecting-unit control line; and a plurality of interconnecting units, wherein: each one of the plurality of interconnecting units is stacked between and connected to each of the first tab portion of one of the plurality of busbars and the second tab portion of another one of the plurality of busbars and separately connected to an interconnecting-unit control line, each one of the plurality of interconnecting units is configured to electrically disconnect the first tab portion and the second tab portion before receiving an electrical input through the interconnecting-unit control line, and each one of the plurality of interconnecting units is configured to electrically connect the first tab portion and the second tab portion after receiving the electrical input through the interconnecting-unit control line.
2. The multilayered flexible interconnect circuit of claim 1, wherein each one of the plurality of interconnecting units is a transistor comprising: a source electrically interconnected with either the first tab portion of the first busbar and the second tab portion of the second busbar, a drain electrically interconnected with the other of either the first tab portion of the first busbar and the second tab portion of the second busbar, and a gate electrically interconnected with the interconnecting-unit control line, wherein: each transistor is configured to reversibly electrically connect the first busbar and the second busbar after receiving the electrical input through the interconnecting-unit control line.
3. The multilayered flexible interconnect circuit of claim 1, wherein each one of the plurality of interconnecting units is a normally-open relay comprising: a first pole electrically interconnected with either the first tab portion of the first busbar and the second tab portion of the second busbar, a second pole electrically interconnected with the other of either the first tab portion of the first busbar and the second tab portion of the second busbar, and an actuation lead electrically interconnected with the interconnecting-unit control line, wherein: each normally-open relay is configured to reversibly electrically connect the first busbar and the second busbar after receiving

the electrical input through the interconnecting-unit control line.

4. The multilayered flexible interconnect circuit of claim 1, wherein each one of the plurality of interconnecting units is a heat-actuated interconnector comprising: a first connecting portion electrically interconnected with either the first tab portion of the first busbar and the second tab portion of the second busbar, a second connecting portion electrically interconnected with the other of either the first tab portion of the first busbar and the second tab portion of the second busbar, a meltable portion positioned between the first connecting portion and the second connecting portion, and a heating element electrically interconnected with the interconnecting-unit control line, wherein: the heating element is configured to melt the meltable portion after receiving the electrical input through the interconnecting-unit control line and thereby irreversibly electrically connect the first busbar and the second busbar.

5. The multilayered flexible interconnect circuit of claim 4, wherein the heating element is positioned between the first connecting portion and the second connecting portion.

6. The multilayered flexible interconnect circuit of claim 4, wherein the heating element is positioned either on an opposite side of the first connecting portion from the second connecting portion or on an opposite side of the second connecting portion from the first connecting portion.

7. The multilayered flexible interconnect circuit of claim 1, wherein: each one of the plurality of busbars comprises a first conductive layer and a second conductive layer, the contact portion is formed by both the first conductive layer and the second conductive layer, stacked in the contact portion, the first tab portion is formed by the first conductive layer such that the second conductive layer does not extend to the first tab portion, and the second tab portion is formed by the second conductive layer such that the first conductive layer does not extend to the second tab portion.

8. The multilayered flexible interconnect circuit of claim 7, wherein the first conductive layer and second conductive layer are separated from each other in the contact portion by an insulation layer.

9. The multilayered flexible interconnect circuit of claim 7, wherein the first conductive layer and second conductive layer directly interface each other in the contact portion.

10. The multilayered flexible interconnect circuit of claim 9, wherein the first conductive layer and the second conductive layer directly interface each other and are welded to one another in the contact portion.

11. The multilayered flexible interconnect circuit of claim 1, wherein at least one of the plurality of interconnecting units reversibly electrically connects the first tab portion and the second tab portion after receiving the electrical input.

12. The multilayered flexible interconnect circuit of claim 1, wherein at least one of the plurality of interconnecting units is connected to one of the plurality of busbars connected to a terminal of a battery and another one of the plurality of busbars connected to another terminal of the battery.

13. The multilayered flexible interconnect circuit of claim 1, wherein at least one of the plurality of interconnecting units is connected to one of the plurality of busbars connected to a terminal of a battery and another one of the plurality of busbars connected to another terminal of another battery.

14. The multilayered flexible interconnect circuit of claim 1, wherein each of the plurality of interconnecting units irreversibly electrically connects the first tab portion and the second tab portion after receiving the electrical input.

15. The multilayered flexible interconnect circuit of claim 1, wherein the electrical input has a voltage of 1-5 volts.

16. A battery assembly comprising: a plurality of busbars comprising a first busbar and a second busbar; a plurality of interconnecting units, each connected to two of the plurality of busbars and separately connected to an interconnecting-unit control line; a plurality of battery cells comprising cell terminals; and a battery management system, wherein: each of the plurality of busbars comprises a first conductive layer and a second conductive layer, each of the plurality of busbars is defined by a contact portion, a first tab portion, and a second tab portion, the contact portion is formed by both the first conductive layer and the second conductive layer, stacked in the contact

portion, the first tab portion is formed by the first conductive layer such that the second conductive layer does not extend to the first tab portion, the second tab portion is formed by the second conductive layer such that the first conductive layer does not extend to the second tab portion, the cell terminals of each one of the plurality of battery cells are each electrically connected to different ones of the plurality of busbars, each of the interconnecting-unit control lines is electronically connected with the battery management system, the plurality of interconnecting units comprise an interconnecting unit stacked between and connected to each of the first tab portion of the first busbar and the second tab portion of the second busbar, each one of the plurality of interconnecting units maintains the two of the plurality of busbars electrically disconnected from each other before receiving an electrical input through the interconnecting-unit control line, and each one of the plurality of interconnecting units is configured to electrically connect the two of the plurality of busbars after receiving the electrical input through the interconnecting-unit control line.

17. The battery assembly of claim 16, further comprising a plurality of voltage traces, wherein each of the voltage traces electrically connects one of the plurality of busbars to the battery management system.

18. The battery assembly of claim 16, further comprising a battery management system power source electrically connected to the Battery Management System, wherein the battery management system power source is at least one of the plurality of battery cells.

19. A method of operating a battery assembly, the method comprising: providing a battery pack comprising: a plurality of busbars comprising a first busbar and a second busbar; a plurality of interconnecting units, each connected to two of the plurality of busbars and separately connected to an interconnecting-unit control line; a plurality of battery cells comprising cell terminals; a plurality of voltage traces; and a battery management system, wherein: each of the plurality of busbars comprises a first conductive layer and a second conductive layer, each of the plurality of busbars is defined by a contact portion, a first tab portion, and a second tab portion, the contact portion is formed by both the first conductive layer and the second conductive layer, stacked in the contact portion, the first tab portion is formed by the first conductive layer such that the second conductive layer does not extend to the first tab portion, the second tab portion is formed by the second conductive layer such that the first conductive layer does not extend to the second tab portion, the cell terminals of each one of the plurality of battery cells are each electrically connected to different ones of the plurality of busbars, each of the interconnecting-unit control lines is electronically connected with the battery management system, each of the voltage traces electrically connects one of the plurality of busbars to the battery management system, the battery management system is configured to determine SOH of each of the plurality of battery cells by measuring voltages at each of the plurality of busbars, the plurality of interconnecting units comprise an interconnecting unit stacked between and connected to each of the first tab portion of the first busbar and the second tab portion of the second busbar, each one of the plurality of interconnecting units maintains the two of the plurality of busbars electrically disconnected from each other before receiving an electrical input through the interconnecting-unit control line, and each one of the plurality of interconnecting units is configured to electrically connect the two of the plurality of busbars after receiving the electrical input through the interconnecting-unit control line; determining at the battery management system if the SOH of any one of the plurality of battery cells is unsatisfactory by comparing voltages measured at each of the plurality of busbars to a predetermined list of voltages; selecting at the battery management system which of the plurality of interconnecting units to provide an electrical input to; and providing, by the battery management system, an electrical input to the selected ones of the plurality of interconnecting units via at least one of the interconnecting-unit control line.

20. The method of operating a battery assembly of claim 19, the method further comprising: determining at the battery management system an SOC of any one of the plurality of battery cells with unsatisfactory SOH.

